BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

COMPUTER SCIENCE

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) (36)

Maximum Marks: 36

Note: This section consists of 36 part questions and all are to be answered. Each question carries ONE marks. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 2 Hours — Maximum Marks: 100 (As per condensed syllabus). OPTION - I - PROGRAMMING USING "C"

Q1. Select the correct answer for each from the given options:

Qi. This data type in C occupies 4 times as many bytes of memory as type char:

(A) int
(B) float
(C) double
(D) long float

Qii. A block of instructions that performs a specific task, is a/an:

(A) function
(B) loop
(C) string
(D) array

Qiii. Every statement in C must end with:

(A) :
(B) }
(C) ;
(D) .

Qiv. A variable that can hold the value 234 having a data type:

(A) char
(B) float
(C) int
(D) double

Qv. A relational operator is used to:

(A) change variable to logical values
(B) compare values
(C) distinguish different types of variable
(D) combine values

Qvi. This statement causes an exit only from the inner most switch:

(A) continue
(B) goto
(C) break
(D) default

Qvii. Multiple increment expression in a for loop expression is separated by:

(A) <
(B) >
(C) ;
(D) ,

Qviii. A string is terminated by a null character which is written as:

(A) \0
(B) //
(C) \n
(D) \\

Qix. Type char can accommodate numbers from:

(A) -32768 to +32767
(B) 3.4e-38 to 3.4e+38
(C) -128 to 127
(D) 0 to 65565

Qx. The function designed specially to read string from the keyboard:

(A) getch()
(B) puts()
(C) gets()
(D) getchar()

Qxi. An external variable is defined:

(A) in the first function that uses it
(B) in any function that uses it
(C) in main () only
(D) outside of any function

Qxii. This function is used to clear the screen:

(A) clear
(B) clrscr
(C) Clrsr
(D) CLS()

Qxiii. This format specifier is used for exponential notation:

(A) %e
(B) %x
(C) %xd
(D) %d

Qxiv. In switch statement if no match is found with any case statement, only the statements following are execute in:

(A) else
(B) continue
(C) default
(D) break

Qxv. A loop performed within another loop is said to be:

(A) nested loop
(B) conditional loop
(C) recursive loop
(D) continuous loop

Qxvi. Three types of control structures are:

(A) initialization, condition, increment
(B) sequence, selection, iteration
(C) sequential, selection, conditional
(D) initialization, condition, decrement

Qxvii. String is:

(A) a collection of characters
(B) a list of characters
(C) an array of characters
(D) a single character

Qxviii. This will be output of 22% 3:

(A) 3
(B) 1
(C) 22
(D) 0

Qxix. The function scanf() reads:

(A) any possible number
(B) a single character
(C) any possible variable type
(D) a character and a string

Qxx. This loop will be executed at least once whether its condition is false:

(A) for
(B) while
(C) do-while
(D) continuous loop

Qxxi. A variable which can be accessed by all program modules is called:

(A) global variable
(B) local variable
(C) argument
(D) private variable

Qxxii. Case statement is used with:

(A) printf()
(B) scanf()
(C) switch ()
(D) getch()

Qxxiii. This statement is incorrect in C:

(A) int a;
(B) float b;
(C) double c;
(D) character c;

Qxxiv. The escape sequence is used for:

(A) tab
(B) backspace
(C) end of string
(D) new line

Qxxv. This is called the address operator:

(A) &
(B) %
(C) &&
(D) |

Qxxvi. The library function sqrt (1024) returns:

(A) 512
(B) 32
(C) 256
(D) 64

Qxxvii. Quantities whose values cannot be changed during program execution are:

(A) data files
(B) variables
(C) constants
(D) functions

Qxxviii. If a = -5, the statement a + = 5 returns:

(A) -15
(B) 0
(C) 10
(D) -5

Qxxix. This expression is valid:

(A) a++1
(B) 6++
(C) a++
(D) a++1

Qxxx. This operator operates on only one operand:

(A) =
(B) = =
(C) &&
(D) !

Qxxxi. This is invalid variable declaration

(A) double c;
(B) unsigned int k;
(C) float p,q
(D) char name;

Qxxxii. What will be the output of the following statement: Printf("%.2f", 633.6658)

(A) 63.33
(B) 63
(C) 633.66
(D) 633.6

Qxxxiii. This statement can terminate any loop immediately:

(A) continue
(B) switch
(C) break
(D) goto

Qxxxiv. A format specifier used for int type variable is:

(A) %c
(B) %f
(C) %d
(D) %s

Qxxxv. Function prototype is called:

(A) Function definition
(B) Function Declaration
(C) Function Calling
(D) Function Repetation

Qxxxvi. This sub directory contains header files:

(A) lib
(B) bin
(C) include
(D) bgi

Section 'B' (Short-Answer Questions) (25)

Maximum Marks: 25

Note: Answer any five part questions from this section. All questions carry equal marks. Answer should not exceed 3 to 4 lines.

Q2.

(i) What is IDE? Write few shortcut keys and their functions in IDE.

(ii) Write a program to print the factorial of an inputted number.

(iii) What is meant by Escape Sequence? Define any four with examples.

(iv) Convert the following mathematical expressions into equivalent 'C' expressions. (Attempt any two) a) $B = (x + y)^2 - (a^2 + b^2)$ b) $p = (a^2 - b^2)/(a - b^2)$ c) $y = (a + b)^n$ d) $X = \sqrt{(b + a^2)}$

(v) What is main difference between Source code and Object code.

(vi) Define Logical operators with examples.

(vii) What are the library functions? Name any four.

(viii) What are Local and Global Variables also give examples:

(ix) By using nested loop write the program to print the following output. 2 4 6 8 10 / 3 6 9 12 15 / 4 8 12 16 20 / 5 10 15 20 25

(x) Write a program to calculate the area of rectangle (Hint A = w*l).

Section "C" (Detailed Answer Questions) (14)

Maximum Marks: 14

Note: Attempt any Two questions from this section. All questions carry equal marks.

Q3.

Explain all data types in C – Language

Q4.

How many types of decision structures are there in 'C'? Define with examples.

Q5.

What is Function? What are the advantages of using function?

Q6.

Define switch and break Statement with example.

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